

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S5 (PT) (R,S) Examination November 2024 (2019 Scheme)

Course Code: EET435**Course Name: RENEWABLE ENERGY SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Write any three advantages and three disadvantages of conventional energy sources. | (3) |
| 2 | Explain the principle of solar energy conversion into heat energy. | (3) |
| 3 | List and explain any three site selection criteria for OTEC systems. | (3) |
| 4 | What are the main parts of the tidal power plant? Explain. | (3) |
| 5 | Explain the terms -Yaw control, Nacelle, and Wind vane - related to wind energy systems. | (3) |
| 6 | Differentiate the constructional and operational aspects of the horizontal and vertical axis wind turbines. | (3) |
| 7 | Describe the importance of seeding, retention time, and feeding rate for biogas production. | (3) |
| 8 | What is meant by producer gas? How is it produced from biomass? | (3) |
| 9 | What are the different classifications of small hydro power plants? | (3) |
| 10 | List the advantages of fuel cell power plants. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Draw the equivalent circuit of a practical solar cell. Define short circuit current, open circuit voltage, fill factor, conversion efficiency | (10) |
| | b) Compare flat plate collectors and focussing collectors. | (4) |

OR

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|----|---|------|
| 12 | a) What are the major components of stand-alone solar PV systems? Explain the function of each with the help of block diagrams. | (10) |
| | b) Explain the effect of insolation on solar cell performance with the help of I-V and P-V characteristics. | (4) |

Module II

- 13 a) Draw the block diagram and explain the operation of a hybrid-type OTEC system. (10)
- b) What are the limitations of OTEC systems? (4)

OR

- 14 a) Draw a schematic diagram of a double-basin Tidal power plant and explain the working. (10)
- b) What is the principle of tidal power generation? Explain. (4)

Module III

- 15 a) Derive the relation between upstream and downstream velocity of wind to extract maximum power from the wind turbine. (10)
- b) If the wind speed is 20m/s and the blade length is 50m, calculate the power in the wind. Given air density (1.23kg/m). (4)

OR

- 16 a) With the help of neat diagrams explain the constructional details of Darrieus type and multi-bladed type wind turbines. What are their applications? (10)
- b) What are the limitations of wind energy conversion systems? (4)

Module IV

- 17 a) What are the different methods for biomass-fuel conversion? Explain. (10)
- b) What are the different steps involved in the production of biogas? (4)

OR

- 18 a) With the help of block diagrams explain municipal solid waste-based power generation. (10)
- b) Explain the different stages in the biomass gasification process. (4)

Module V

- 19 a) With the help of a neat diagram explain the different types of reaction turbines used in small hydropower plants. (6)
- b) Draw and explain the layout of a small hydel power plant. (8)

OR

- 20 a) Explain the construction and working features of a PEM fuel cell. (8)
- b) Explain the steam reformation method of hydrogen production. (6)
